

AUBURN, CA, USA

WMO: 720267

Lat: 38.955N

Lon: 121.082W

Elev: 1531

StdP: 13.90

Time Zone: -8.00 (NAP)

Period: 04-19

WBAN: 23224

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 32.0	(c) 34.1	(d) 15.8	(e) 13.0	(f) 48.7	(g) 19.1	(h) 15.2	(i) 47.0	(j) 19.7	(k) 47.4	(l) 17.7	(m) 48.6	(n) 2.5	(o) 100	(p) 0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	98.8	66.9	95.0	65.2	92.5	64.6	69.3	92.1	67.9	89.8	66.4	87.8	6.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	86.2	76.9	59.4	80.0	76.4	57.2	73.9	73.4	34.2	91.6	32.9	90.1	31.7	87.7	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.6	(b) 12.6	(c) 11.0	(e) 27.2	(f) 103.9	(g) 3.5	(h) 6.4	(i) 24.6	(j) 108.5	(k) 22.6	(l) 112.2	(m) 20.6	(n) 115.8	(o) 18.0	(p) 120.4		
(a) 15.6	(b) 12.6	(c) 11.0	(e) 25.3	(f) 72.4	(g) 2.9	(h) 3.4	(i) 23.1	(j) 74.8	(k) 21.4	(l) 76.7	(m) 19.8	(n) 78.6	(o) 17.6	(p) 81.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	47.3	49.0	52.4	55.8	63.1	71.8	78.8	77.6	72.9	62.7	53.3	46.1
	DBStd	13.19	5.45	6.23	6.49	7.24	7.39	7.81	5.66	5.40	7.39	6.90	6.42	5.08
	HDD50	456	117	86	49	22	2	0	0	0	0	1	37	142
	HDD65	2916	547	449	393	291	125	26	0	0	19	126	353	587
	CDD50	4462	35	58	123	195	408	655	892	857	687	396	136	20
	CDD65	1447	0	0	2	15	67	231	427	392	256	55	2	0
	CDH74	16844	1	7	16	139	751	2843	5287	4567	2705	508	20	0
	CDH80	8098	0	1	1	20	226	1317	2813	2318	1274	127	1	0
(14) Wind	WSAvg	5.0	4.2	5.4	5.7	5.7	5.5	5.2	5.3	5.0	4.9	4.3	4.0	4.3
(15) Precipitation	PrecAvg	35.4	6.2	5.2	5.7	2.7	0.7	0.3	0.0	0.2	0.7	2.3	5.9	5.4
	PrecMax	64.9	17.0	17.6	15.3	7.7	2.8	1.1	0.9	1.6	3.7	13.9	13.5	14.2
	PrecMin	11.8	0.5	0.3	1.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.0
	PrecStd	11.9	4.3	4.1	3.6	2.3	0.7	0.3	0.2	0.4	1.1	2.8	3.8	3.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	71.9	75.4	82.2	90.6	99.8	102.1	100.1	98.8	89.8	77.0	64.2
		MCWB	51.3	56.1	58.7	62.8	65.9	68.4	69.6	66.1	65.1	61.6	57.8	49.8
	2%	DB	64.0	66.5	72.1	78.9	84.5	95.4	98.8	97.1	93.1	82.4	72.8	61.1
		MCWB	49.9	53.6	56.7	60.8	62.9	66.6	67.5	65.0	62.8	59.2	56.0	49.6
	5%	DB	61.2	63.6	68.3	73.1	81.8	91.0	95.2	93.1	90.7	79.5	69.8	57.1
		MCWB	49.5	51.2	54.8	57.6	61.2	64.6	65.6	64.6	62.5	57.3	54.6	48.7
	10%	DB	56.7	60.6	64.0	69.8	78.7	88.1	91.5	90.7	88.2	75.3	63.8	54.7
		MCWB	48.3	50.0	52.5	56.5	59.9	63.4	65.1	63.7	61.9	56.4	52.3	48.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.4	58.0	60.3	65.3	67.8	71.8	72.8	68.8	67.0	64.2	62.4	56.7
		MCDB	59.9	63.6	72.9	79.4	85.7	94.4	97.8	93.3	86.1	78.6	72.4	58.3
	2%	WB	53.7	55.5	57.8	62.1	65.3	69.2	69.5	67.4	65.2	62.0	58.8	54.3
		MCDB	58.6	63.9	68.8	75.6	83.3	89.9	93.7	91.3	88.1	78.8	67.6	55.8
	5%	WB	51.9	53.5	55.7	59.3	63.0	67.1	68.2	66.1	63.8	60.1	56.4	52.4
		MCDB	56.2	59.8	65.1	71.1	77.9	87.0	91.4	89.8	87.0	74.5	64.4	54.3
	10%	WB	50.1	51.8	53.8	57.1	61.0	65.2	66.6	64.9	62.3	58.2	54.3	50.1
		MCDB	55.8	57.4	62.0	67.4	75.3	83.4	88.7	87.8	84.8	71.0	61.2	54.1
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	16.1	16.6	18.4	21.0	24.1	24.5	24.3	23.2	20.4	17.1	15.1
		MCDBR	22.0	22.3	21.8	22.8	24.6	26.9	25.7	25.0	24.9	24.0	22.9	19.0
		MCWBR	13.1	14.0	13.3	13.4	12.0	11.0	10.1	10.4	11.2	11.3	12.3	11.6
	5% WB	MCDBR	17.4	18.6	19.3	20.9	23.0	24.8	24.2	24.0	23.7	20.5	19.3	12.3
		MCWBR	11.8	12.5	12.3	12.9	11.7	10.8	9.9	10.3	10.9	10.5	11.7	9.2
(40) Clear-Sky Solar Irradiance	taub		0.309	0.310	0.338	0.342	0.357	0.349	0.348	0.332	0.341	0.332	0.310	0.323
	taud		2.514	2.525	2.433	2.424	2.404	2.441	2.443	2.478	2.459	2.483	2.525	2.450
	Ebn at Noon		276	292	292	296	292	293	293	296	287	279	273	259
	Edh at Noon		24	27	33	36	37	36	36	33	32	28	24	24
(44) All-Sky Solar Radiation	RadAvg		702	997	1417	1881	2300	2622	2616	2347	1892	1327	843	613
	RadStd		119	138	167	181	165	121	91	81	72	96	67	105

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	+0.72	N/A	-1.96	N/A	N/A	N/A	N/A	-205	+248	+76

Nomenclature: See separate page